Improving microsurgical suture training with automated phase recognition and skill assessment via deep learning.

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Abstract

Microsurgical suturing demands a high level of precision, skill, and extensive training to ensure success in delicate procedures. In this study, we created a deep-learning approach for automating phase recognition and skill assessment in microsurgical suturing. We processed and segmented microsurgical videos using three variants of modified Long-Range Recurrent Convolutional Networks (LRCNs) in order to classify phases and evaluate surgeon performance. Data augmentation techniques were applied to address the class imbalance, and a skipping window strategy was used to select representative frames. The models were tested on a dataset of novice and expert surgeons. Our findings revealed that the models could reliably distinguish between skill levels by analyzing confidence vs the time spent in each phase. We also highlight challenges in identifying phases with overlapping visual and temporal features; however, the models demonstrated good generalization capability across other datasets. The proposed approach shows promise in improving surgical training and skill assessments, potentially leading to better surgical results and more individualized training programs.

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